GURMOHIT SINGH

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EDUCATION

The University of Texas at Austin - GPA: 3.95 | ***Bachelor of Science, Computer Science*** | **Aug 2024 – May 2027 (Expected)**
Lone Star Community College - **GPA:** 3.98 | ***Associate of Science*** | **Aug 2021 – May 2024**

SKILLS

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript, C#
ML/Data: pandas, NumPy, scikit-learn, LightGBM, Matplotlib, TensorFlow, PyTorch, feature engineering, cross-validation, hyperparameter tuning, reinforcement learning, Bayesian networks, neural networks
Systems/Tools: Git, Linux, GPU-Z telemetry, performance benchmarking, Qiskit, IBM Quantum, PySide6, Excel
Web Development: React, Next.js, Vite, Tailwind CSS, Spring Boot, REST APIs, HTML/CSS
Coursework: Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Quantum Computing, Energy-Efficient Computing

EXPERIENCE

WPSPINS – Houston, TX | ***Web Development Intern*** | **Feb – May 2026**
- Developed static and WordPress-based websites for local business clients, translating design/content requirements into responsive, production-ready pages.
- Customized WordPress themes, layouts, navigation, forms, and content sections to support client branding and site functionality.
- Used HTML, CSS, and JavaScript tooling to improve page structure, mobile responsiveness, and maintainability across client sites.

Tandoorish LLC – Houston, TX | ***Web Manager/Developer*** | **Jun 2024 – Present**
- Managed website updates for menus, promotions, events, and online ordering, ensuring customer-facing content stays accurate across business changes.
- Used SEO and Google Analytics insights to improve online visibility, contributing to a 200%+ increase in site traffic since takeover.
- Maintained lightweight web features and coordinated with stakeholders to prioritize changes based on customer experience and operational needs.

Operations Manager | **Nov 2022 – Present**
- Led daily operations, staff training, inventory management, and customer service in a fast-paced, high-volume environment.
- Assisted with food preparation, kitchen coordination, quality control, and menu execution to ensure consistency and efficiency.
- Improved cost control through supplier coordination, budgeting, inventory oversight, and waste reduction initiatives, lowering operational costs by approximately 30%.

PROJECTS

Aviation Damage Risk Predictor | *Python, Pandas, scikit-learn, LightGBM, NumPy* | **Mar – May 2026**
- Built an imbalanced binary classification pipeline to predict indicated aircraft damage from 307K+ aviation incident records.
- Engineered 104 interpretable features from missingness patterns, date/time fields, high-cardinality categorical variables, text keywords, frequency encodings, and domain-specific risk interactions.
- Evaluated LightGBM variants with 5-fold stratified cross-validation, threshold tuning, and ablation testing; achieved 0.939 out-of-fold ROC-AUC and 0.860 balanced accuracy.

Pacman AI Project Suite | *Python, Search, Reinforcement Learning, Bayesian Networks* | **Jan – May 2026**
- Implemented AI agents across five Pacman projects covering graph search, adversarial search, value iteration, Q-learning, Bayesian network inference, and supervised classification.
- Applied these methods to maze navigation, multi-agent decision-making, reinforcement learning, probabilistic reasoning, and digit/Pacman behavior classification.

Energy-Efficient Graphics API Benchmark | *Vulkan, Direct3D 12, OpenGL, GPU-Z* | **Feb – May 2026**
- Conducted a controlled GPU efficiency study across Vulkan, Direct3D 12, and OpenGL APIs with various graphical workloads.
- Collected benchmark scores, FPS, and power/thermal metrics across repeated trials, then derived performance-per-watt metrics
- Found Vulkan delivered the best measured performance per watt in both benchmarks, including 0.727 FPS/W in GravityMark, versus 0.641 FPS/W for OpenGL and 0.633 FPS/W for Direct3D 12, at roughly equal board power.

QARDS – THE Quantum Card Game | *Python, Qiskit, PySide6 (Qt)* | **Sep – Oct 2025**
- Built a digital card game that models quantum concepts (superposition, entanglement, interference) as core mechanics.
- Implemented interference-based mechanics inspired by Grover's diffusion operator.
- Delivered both a CLI prototype and a GUI application, earning **2nd place** at UT IBM Qiskit Fall Fest Hackathon (2025).

ACTIVITIES

Summer 2025 Quantum Optics Program - Applied Research Laboratories, UT Austin – Austin, TX | **Jun 2025 – Aug 2025**
- Gained hands-on experience with quantum computing algorithms and their implementations with optical lab equipment.
- Implemented algorithms such as Quantum Teleportation, Entanglement Swapping, and a violation of the Bell-CHSH Inequality.

UT Quantum Collective – Austin, TX | **Aug 2025 – Present**
- Participate in quantum computing labs, reading groups, hackathons, and Qiskit/IBM Quantum circuit-building initiatives.